

VPBank Technology Hackathon 2025

General Brief

Please fill up this table and use this document as a template to write your proposal.

Challenge Statement	Challenge #5: AI/ML ENHANCED CREDIT SCORING
Team Name	Nexus Score

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Solutions Introduction

Our solution implements a **3-pipeline architecture** for AI-enhanced credit scoring:

1. **Pipeline 1:** Batch Data Enrichment Pipeline - Transform raw data into ML-ready features
2. **Pipeline 2:** MLOps Training Pipeline - Automated model training and validation
3. **Pipeline 3:** Real-time Inference Pipeline - Live credit scoring with explanations

Core Technologies:

- AWS Step Functions for workflow orchestration
- Amazon Textract for document processing
- Amazon Bedrock (Claude) for AI analysis
- SageMaker for ML operations
- XGBoost ensemble models

Data Inputs Assumed:

1. **Structured Data Schema (27 features)** from [Kaggle](#)
2. **Unstructured Data:**
 - PDF contracts, bank statements, utility bills
 - Images of documents, property photos
 - Alternative data sources (videos, social media,...)

Impact of Solution

How does your solution benefit the society / the target audience?

Why is your solution a good solution? How is it better than existing solutions in the market / competitors?

What is your solution's competitive advantage / unique selling point?

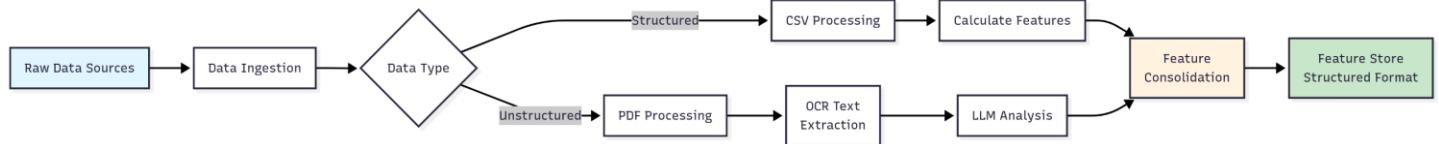
Deep Dive into Solution

Our solution implements a three-pipeline architecture that transforms raw customer data into accurate, explainable credit scores through automated AI processing.

Pipeline 1: Data Enrichment Pipeline

Objective: Automatically transform raw data into structured Feature Store ready for training.

Flow Diagram:



Main Processing Flow:

1. Data Ingestion:

- Multi-format support: CSV transaction files, PDF contracts, images, JSON data
- Real-time streaming and batch processing capabilities
- Data validation and quality checks at entry point
- Automatic file format detection and routing

2. Parallel Processing: Process two data types simultaneously:

○ Structured Data Processing:

- Extract features from CSV, such as: avg_monthly_income, income_volatility, late_payments,...
- Calculate derived metrics: debt-to-income ratio, spending patterns, account aging
- Handle missing values using statistical imputation
- Apply data normalization and standardization

○ Unstructured Data Processing:

- OCR technology for text extraction from PDFs/images
- Support multiple languages and document formats
- Handle poor quality scans and handwritten text
- Extract structured information: dates, amounts, signatures

3. AI Enrichment:

○ LLM Analysis Pipeline:

- Custom prompts for financial document analysis
- Extract: contract duration, risk indicators, termination clauses, guarantees
- Sentiment analysis of customer communications
- Output format: structured JSON with confidence scores

○ Feature Engineering:

- Create composite risk scores from multiple sources
- Time-series feature extraction for behavioral patterns

- Cross-validation of extracted information

4. Feature Consolidation:

- Merge structured and unstructured features
- Feature selection using statistical methods
- Handle feature conflicts and inconsistencies
- Create final feature vector with 50+ dimensions

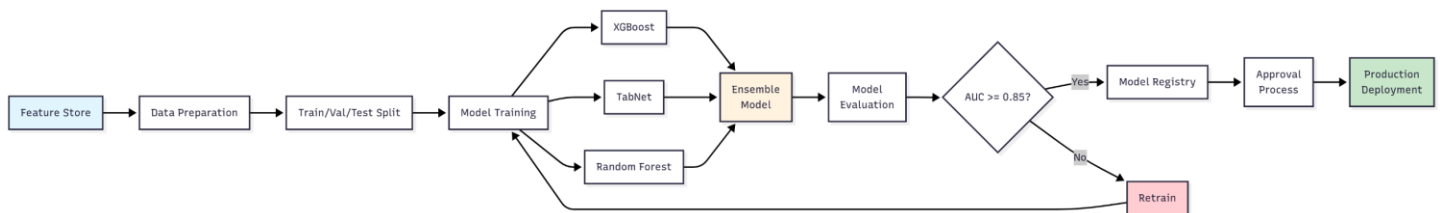
5. Storage:

- Versioned feature store with audit trail
- Partitioned by customer segments and time periods
- Real-time and batch access patterns supported
- Data lineage tracking for regulatory compliance

Pipeline 2: ML Training Pipeline

Objective: Automate model training, evaluation, and deployment workflow.

Flow Chart:



Implementation Steps:

1. Data Preparation:

- Load feature-engineered data from Feature Store
- Advanced train/validation/test split with stratification
- Handle temporal dependencies for time-series features
- Apply data balancing techniques (SMOTE-ENN for imbalanced classes)
- Feature scaling and normalization pipeline

2. Model Training Architecture:

- Ensemble Approach:
 - XGBoost: Gradient boosting with L1/L2 regularization, handles missing values
 - TabNet: Deep learning for tabular data with sequential attention mechanism
 - Random Forest: Bagging ensemble with feature importance ranking
- Hyperparameter Optimization:

- Bayesian optimization for efficient parameter search
- Cross-validation with temporal splits for time-series data
- Early stopping to prevent overfitting
- Training Infrastructure:
 - Distributed training for large datasets
 - GPU acceleration for deep learning components
 - Automated MLOps pipeline with experiment tracking

3. Model Evaluation & Validation:

- Performance Metrics:
 - Primary: AUC-ROC, AUC-PR for imbalanced datasets
 - Secondary: Precision, Recall, F1-Score by risk categories
 - Business metrics: Expected loss, approval rates
- Advanced Analysis:
 - SHAP values for global and local feature importance
 - Permutation importance for feature selection
 - Bias detection across demographic groups
 - Stability testing with out-of-time validation
- Model Interpretability:
 - Generate decision trees for rule extraction
 - Feature interaction analysis
 - Counterfactual explanations

4. Model Registry & Governance:

- Automated Registration:
 - Models with AUC ≥ 0.85 and bias metrics within threshold
 - Comprehensive metadata: training data, hyperparameters, performance
 - A/B testing framework for model comparison
- Version Control:
 - Git-based model versioning
 - Rollback capabilities for production models
 - Change tracking and approval workflows
- Compliance:
 - Regulatory documentation generation

- Fairness and bias auditing
- Model risk management documentation

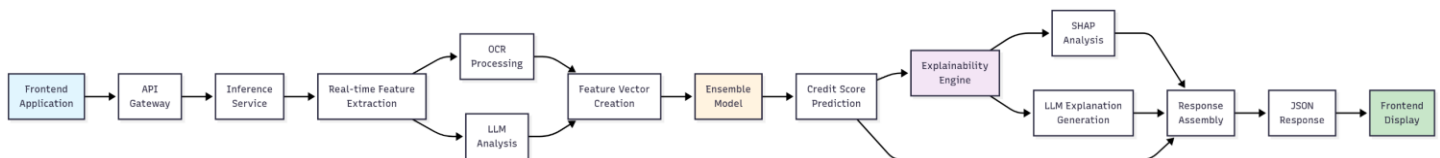
5. Deployment Pipeline:

- Conditional Deployment:
 - Performance thresholds and business rule validation
 - Canary releases with gradual traffic shifting
 - Real-time monitoring and automatic rollback
- Infrastructure:
 - Containerized deployment with auto-scaling
 - Load balancing and failover mechanisms
 - Multi-region deployment for disaster recovery

Pipeline 3: Real-time Inference Pipeline

Objective: Provide API endpoint for real-time credit scoring with explainability.

Flow Chart:



System Architecture:

1. API Gateway & Load Management:

- RESTful API with OpenAPI specification
- Rate limiting and throttling mechanisms
- Authentication and authorization (OAuth 2.0)
- Request validation and sanitization
- Support for both synchronous and asynchronous processing

2. Real-time Feature Extraction Engine:

- Document Processing:
 - Multi-format OCR with confidence scoring
 - Image preprocessing and enhancement
 - Text extraction with layout preservation
 - Language detection and translation capabilities
- LLM Analysis Pipeline:

- Prompt engineering for financial document types
- Named entity recognition for financial terms
- Risk indicator extraction with confidence levels
- Structured output validation and post-processing
- Feature Vector Assembly:
 - Real-time feature calculation matching training pipeline
 - Feature drift detection and alerting
 - Data quality checks and anomaly detection
 - Caching mechanisms for performance optimization

3. **Model Inference Service:**

- Ensemble Prediction:
 - Load balanced calls to XGBoost, TabNet, and Random Forest
 - Weighted voting with dynamic weight adjustment
 - Confidence intervals and prediction uncertainty
 - Model health monitoring and automatic failover
- Performance Optimization:
 - Model caching and prediction memoization
 - Batch prediction for efficiency
 - GPU acceleration for deep learning models
 - Sub-100ms response time target

4. **Explainability Engine:**

- Feature Importance Analysis:
 - Real-time SHAP value calculation
 - Local feature importance for individual predictions
 - Comparative analysis against peer groups
 - Risk factor ranking and categorization
- Natural Language Generation:
 - Template-based explanations for consistency
 - LLM-powered narrative generation
 - Multi-language support for explanations
 - Regulatory-compliant explanation formats
- Visualization Components:

- Interactive charts for risk factors
- Decision tree visualization
- Feature contribution plots
- Historical trend analysis

5. **Response Assembly & Delivery:**

- Comprehensive Output:
 - Credit score (0-850 scale) with confidence interval
 - Risk category (Low/Medium/High) with probability distribution
 - Top 5 contributing factors with impact scores
 - Natural language explanation (150-300 words)
 - Recommended actions and next steps
- Quality Assurance:
 - Response validation and consistency checks
 - Latency monitoring and optimization
 - Error handling and graceful degradation
 - Audit logging for regulatory compliance
- Integration Support:
 - Multiple response formats (JSON, XML)
 - Webhook callbacks for asynchronous processing
 - SDKs for popular programming languages
 - Real-time streaming capabilities

Architecture of Solution

Our AI-Enhanced Credit Scoring solution leverages AWS's comprehensive cloud ecosystem to create a seamless, scalable, and secure platform. Rather than simply deploying individual services, we've architected an integrated solution where AWS services work in concert to deliver end-to-end automation, real-time processing, and enterprise-grade reliability:

Pipeline 1: Data Enrichment using AWS Event-Driven Architecture

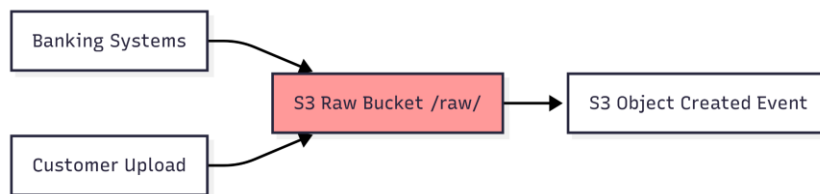
AWS Service Integration Strategy:

- **S3 as Event Source:** Raw data upload to S3 /raw/ folder automatically triggers Step Functions workflow through S3 event notifications, enabling real-time processing without polling.

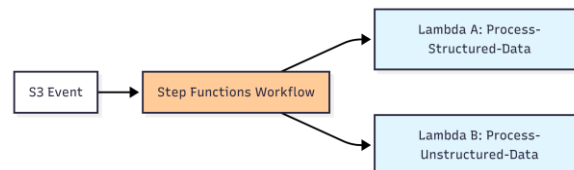
- **Step Functions Orchestration:** Coordinates parallel Lambda executions for structured and unstructured data processing, handling retries and error states. The workflow ensures both processing paths complete before consolidation.
- **Textextract + SNS Integration:** Textextract processes PDFs asynchronously and publishes completion notifications to SNS Topic. This decouples text extraction from downstream LLM processing, enabling scalable document processing.
- **Lambda + Bedrock Chain:** Lambda C subscribes to SNS notifications and uses Amazon Bedrock's Claude model with custom prompts to extract structured insights from unstructured text, converting documents into machine-readable features.

Flow Diagram:

1. Data Ingestion Layer



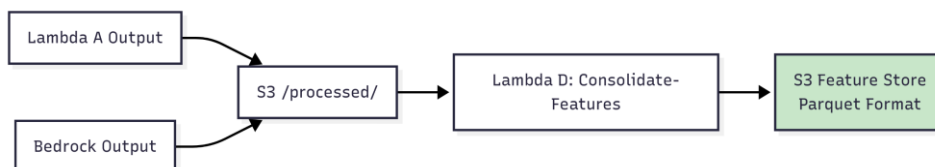
2. Processing Orchestration Layer



3. AI Processing Layer



4. Feature Consolidation Layer



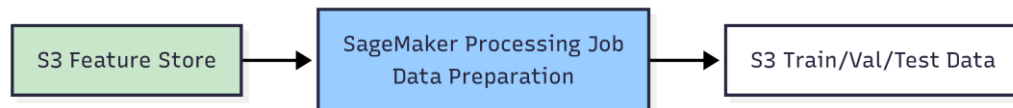
Pipeline 2: MLOps with SageMaker Ecosystem

AWS Service Integration Strategy:

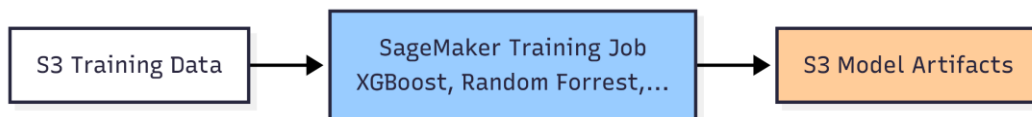
- **SageMaker Processing Jobs:** Handle data preparation and model evaluation using managed compute instances that automatically scale based on data volume and terminate when complete, optimizing costs.
- **SageMaker Training with XGBoost:** Uses AWS's optimized XGBoost algorithm container with automatic hyperparameter tuning and distributed training capabilities for large datasets.
- **S3 Integration Pattern:** All intermediate data (splits, models, evaluations) stored in S3 with versioning enabled, creating complete audit trail for model lineage and regulatory compliance.
- **Conditional Model Registration:** Lambda function evaluates model performance metrics and only registers high-performing models ($AUC \geq 0.85$) in SageMaker Model Registry with "Pending Approval" status.

Flow Diagram:

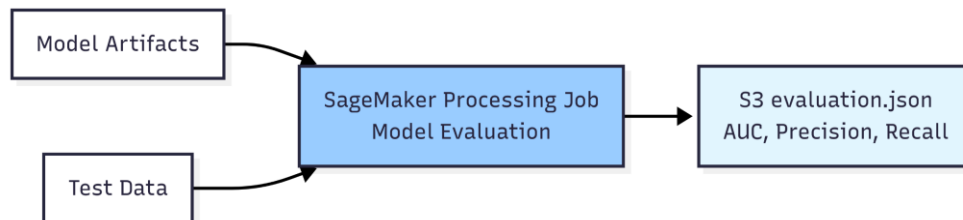
1. Data Preparation Layer



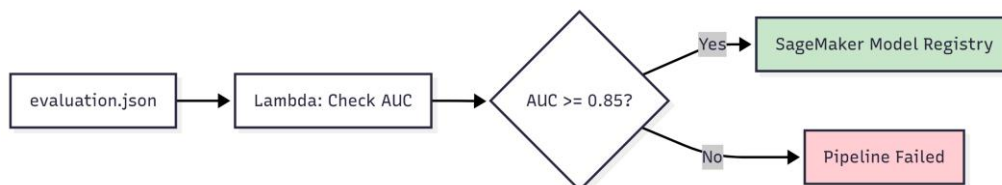
2. Model Training Layer



3. Model Evaluation Layer



4. Model Registry Layer



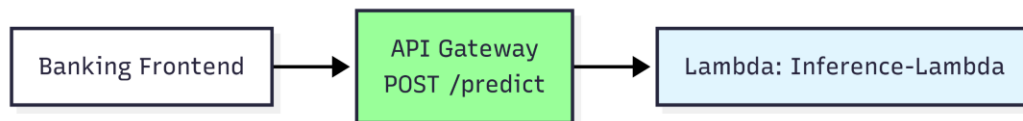
Pipeline 3: Real-time Inference with Serverless Architecture

AWS Service Integration Strategy:

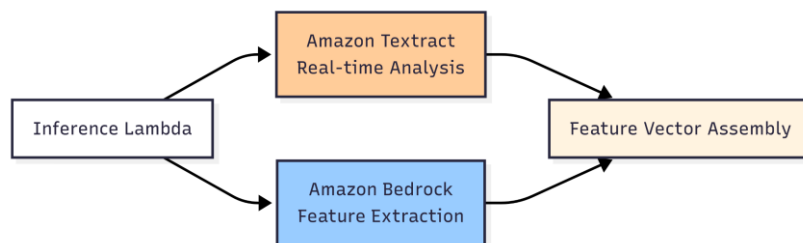
- **API Gateway Integration:** Provides built-in rate limiting, request validation, and authentication. Custom authorizers can integrate with existing bank identity systems.
- **Lambda Real-time Processing:** Performs mini-version of Pipeline 1 processing in real-time, calling Textract and Bedrock synchronously for immediate feature extraction from uploaded documents.
- **SageMaker Serverless Endpoint:** Automatically scales from zero to handle inference requests without managing infrastructure. Provides sub-second response times for real-time scoring requirements.
- **Bedrock for Explanations:** Uses Claude model to generate natural language explanations of credit decisions, taking prediction results and feature importance to create regulatory-compliant explanations

Flow Diagram:

1. API Gateway Layer



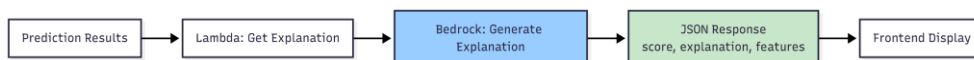
2. Real-time Processing Layer



3. Model Inference Layer

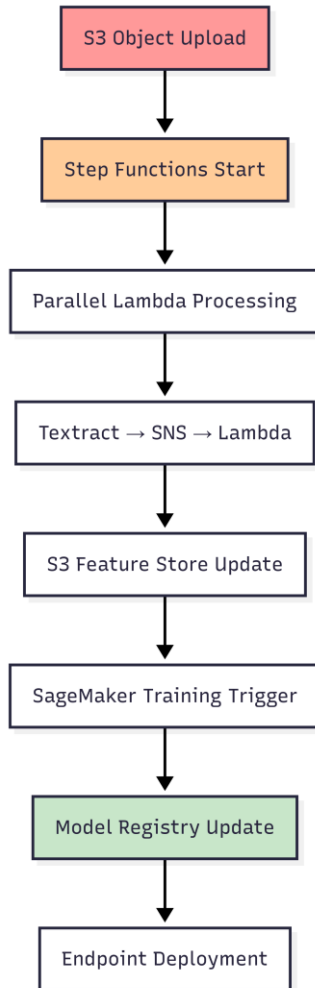


4. Explanation Layer



Cross-Pipeline AWS Integration:

Event-Driven Processing Flow:



Service Communication Strategy:

- **Asynchronous Processing:** S3 events trigger Step Functions, which orchestrate Lambda functions. Textract uses SNS for completion notifications, ensuring scalable document processing without blocking.
- **State Management:** Step Functions maintain processing state across multiple Lambda invocations, handling failures and retries automatically while providing visibility into processing status.
- **Data Flow Optimization:** All data flows through S3 as the central hub, with each service reading from and writing to specific S3 prefixes, creating clear data lineage and enabling easy debugging.
- **Serverless Scaling:** Lambda functions and SageMaker Serverless endpoints automatically scale based on demand, eliminating infrastructure management while maintaining performance.